

WIRELESS NETWORKING

Day – 5 Networking measuring tools.

Task – 1 Testing the connectivity.

- Press Windows Key + R, type cmd, and hit Enter. *(On Mac, open Terminal).*
- Type ping and press Enter.
- Look at the very bottom line for Average = XXms. That number is your average response time.

Task – 2 Using the tracer route command.

- Run tracer in Command Prompt (or traceroute).
- Here are the 4 main types of hops that appear in the output:
- Hop 1 (Local Gateway): Your home WiFi router (e.g., 192.168.1.1 or 192.168.10.1).
- Hops 2–4 (ISP Network): Your Internet provider's local switches and central routers (e.g., Jio, Airtel, or local broadband).
- Hops 5–8 (Internet Backbone): High-speed transit routers moving data across major national networks.
- Final Hops (Flipkart / CDN Edge Server): Flipkart's cloud infrastructure or Content Delivery Network (like Akamai) hosting the site.

Task – 3 Using the nslookup tool.

- Run nslookup [www.zomato.com](<https://www.zomato.com>) in Command Prompt or Terminal.
- **Recorded Result:**
- **Domain:** [www.zomato.com](<https://www.zomato.com>)
- **Resolved IP Address:** 13.224.238.29

Task – 4 Ping checking on the both device.

- When you ping an offline local device:
- You run: `ping 192.168.1.5`
- You get: Destination host unreachable or Request timed out
- Result: 100% packet loss (0 packets received) because no device was active at that address to send a response back.

Task – 5 Giving some reports.

- Network Diagnostics Summary Report
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- (a) Website Not Loading — Traceroute (tracert): Most useful for pinpointing exactly where a path breaks; for example, tracing [\[www.flipkart.com\]\(https://www.flipkart.com\)](https://www.flipkart.com) reveals if data stops at your local router or fails at an intermediate ISP hop.
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- (b) Slow Connection — Ping (ping): Best for measuring network latency and packet loss; for instance, pinging [\[www.youtube.com\]\(https://www.youtube.com\)](https://www.youtube.com) showed an average response time of 16ms to verify connection speed.
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- (c) Suspecting DNS Issues — NSLookup (nslookup): Essential for checking if domain names resolve correctly to IP addresses; for example, running nslookup [\[www.zomato.com\]\(https://www.zomato.com\)](https://www.zomato.com) successfully returned AWS CDN IPs like 13.224.238.29.
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